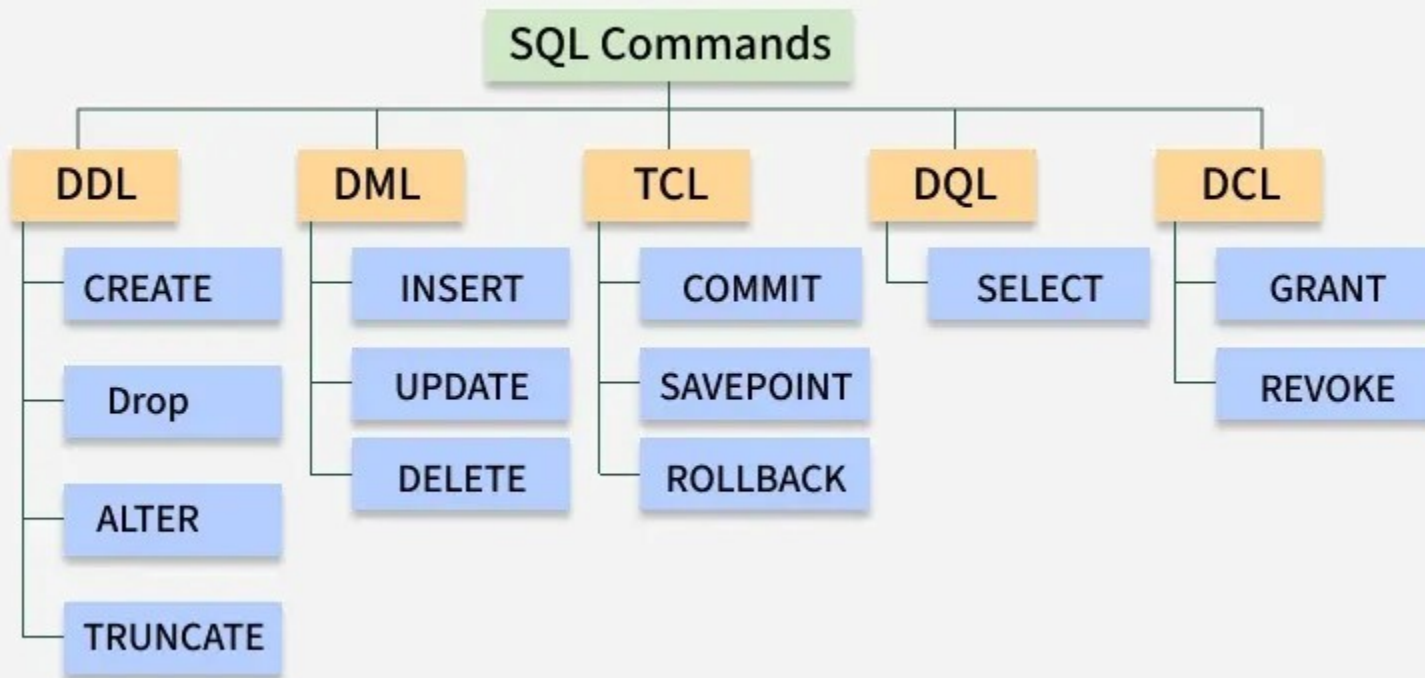


SQL COMPONENTS

SQL commands are fundamental building blocks used to perform given operations on database.

The operations include queries of data, creating a table, adding data to tables, dropping the table, modifying the table and set permission for users.

SQL Commands are mainly categorized into five categories:



DDL - Data Definition Language

DDL (Data Definition Language) consists of SQL commands that can be used for defining, altering and deleting database structures such as tables, indexes and schemas. It simply deals with descriptions of the database schema and is used to create and modify the structure of database objects in the database

Command	Description	Syntax
<u>CREATE</u>	Create database or its objects (table, index, function, views, store procedure and triggers)	CREATE TABLE table_name (column1 data_type, column2 data_type, ...);
<u>DROP</u>	Delete objects from the database	DROP TABLE table_name;
<u>ALTER</u>	Alter the structure of the database	ALTER TABLE table_name ADD COLUMN column_name data_type;
<u>TRUNCATE</u>	Remove all records from a table, including all spaces allocated for the records are removed	TRUNCATE TABLE table_name;
<u>COMMENT</u>	Add comments to the data dictionary	COMMENT ON TABLE table_name IS 'comment_text';
<u>RENAME</u>	Rename an object existing in the database	RENAME TABLE old_table_name TO new_table_name;

Example:

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    hire_date DATE  
);
```

In this example, a new table called employees is created with columns for employee ID, first name, last name and hire date.

2. DML - Data Manipulation Language

DML commands are used to manipulate the data stored in database tables. With DML, you can insert new records, update existing ones, delete unwanted data or retrieve information.

Command	Description	Syntax
<u>INSERT</u>	Insert data into a table	INSERT INTO table_name (column1, column2, ...) VALUES (value1, value2, ...);
<u>UPDATE</u>	Update existing data within a table	UPDATE table_name SET column1 = value1, column2 = value2 WHERE condition;
<u>DELETE</u>	Delete records from a database table	DELETE FROM table_name WHERE condition;

Example:

*INSERT INTO employees (first_name, last_name, department)
VALUES ('Jane', 'Smith', 'HR');*

This query inserts a new record into employees table with first name 'Jane', last name 'Smith' and department 'HR'.

3. DCL - Data Control Language

DCL (Data Control Language) includes commands such as GRANT and REVOKE which mainly deal with the rights, permissions and other controls of the database system. These commands are used to control access to data in the database by granting or revoking permissions.

Command	Description	Syntax
GRANT	Assigns new privileges to a user account, allowing access to specific database objects, actions or functions.	GRANT privilege_type [(column_list)] ON [object_type] object_name TO user [WITH GRANT OPTION];
REVOKE	Removes previously granted privileges from a user account, taking away their access to certain database objects or actions.	REVOKE [GRANT OPTION FOR] privilege_type [(column_list)] ON [object_type] object_name FROM user [CASCADE];

Example:

GRANT SELECT, UPDATE ON employees TO user_name;

This command grants the user user_name the permissions to select and update records in the employees table.

4. TCL - Transaction Control Language

Transactions group a set of tasks into a single execution unit. Each transaction begins with a specific task and ends when all the tasks in the group are successfully completed. If any of the tasks fail, transaction fails. Therefore, a transaction has only two results: success or failure.

Command	Description	Syntax
BEGIN TRANSACTION	Starts a new transaction	BEGIN TRANSACTION [transaction_name];
COMMIT	Saves all changes made during the transaction	COMMIT;
ROLLBACK	Undoes all changes made during the transaction	ROLLBACK;
SAVEPOINT	Creates a savepoint within the current transaction	SAVEPOINT savepoint_name;

Example:

```
BEGIN TRANSACTION;  
UPDATE employees SET department = 'Marketing' WHERE department = 'Sales';  
SAVEPOINT before_update;  
UPDATE employees SET department = 'IT' WHERE department = 'HR';
```

ROLLBACK TO SAVEPOINT before_update;
COMMIT;

In this example, a transaction is started, changes are made and a savepoint is set. If needed, the transaction can be rolled back to the savepoint before being committed.

2. DQL - Data Query Language

DQL is used to fetch data from the database. The main command is SELECT, which retrieves records based on the query. The output is returned as a result set (a temporary table) that can be viewed or used in applications.

Command	Description	Syntax
<u>SELECT</u>	It is used to retrieve data from the database	SELECT column1, column2, ...FROM table_name WHERE condition;
FROM	Indicates the table(s) from which to retrieve data.	SELECT column1 FROM table_name;
<u>WHERE</u>	Filters rows before any grouping or aggregation	SELECT column1 FROM table_name WHERE condition;
<u>GROUP BY</u>	Groups rows that have the same values in specified columns.	SELECT column1, AVG_FUNCTION(column2) FROM table_name GROUP BY column1;
<u>HAVING</u>	Filters the results of GROUP BY	SELECT column1, AVG_FUNCTION(column2) FROM table_name

Command	Description	Syntax
		GROUP BY column1 HAVING condition;
<u>DISTINCT</u>	Removes duplicate rows from the result set	SELECT DISTINCT column1, column2, ... FROM table_name;
<u>ORDER BY</u>	Sorts the result set by one or more columns	SELECT column1 FROM table_name ORDER BY column1 [ASC DESC];
<u>LIMIT</u>	By default, it sorts in ascending order unless specified as DESC	SELECT * FROM table_name LIMIT number;

Note: DQL has only one command, **SELECT**. Other terms like *FROM*, *WHERE*, *GROUP BY*, *HAVING*, *ORDER BY*, *DISTINCT* and *LIMIT* are **clauses** of *SELECT*, not separate commands.

Example:

```
SELECT first_name, last_name, hire_date
FROM employees
WHERE department = 'Sales'
ORDER BY hire_date DESC;
```

This query retrieves employees first and last names, along with their hire dates, from the employees table, specifically for those in the 'Sales' department, sorted by hire date.

3. DML - Data Manipulation Language

DML commands are used to manipulate the data stored in database tables. With DML, you can insert new records, update existing ones, delete unwanted data or retrieve information.

Command	Description	Syntax
<u>INSERT</u>	Insert data into a table	INSERT INTO table_name (column1, column2, ...) VALUES (value1, value2, ...);
<u>UPDATE</u>	Update existing data within a table	UPDATE table_name SET column1 = value1, column2 = value2 WHERE condition;
<u>DELETE</u>	Delete records from a database table	DELETE FROM table_name WHERE condition;

Example:

```
INSERT INTO employees (first_name, last_name, department)
VALUES ('Jane', 'Smith', 'HR');
```

This query inserts a new record into employees table with first name 'Jane', last name 'Smith' and department 'HR'.

4. DCL - Data Control Language

DCL (Data Control Language) includes commands such as GRANT and REVOKE which mainly deal with the rights, permissions and other controls of the database system. These commands are used to control access to data in the database by granting or revoking permissions.

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SAVEPOINT before_update;
UPDATE employees SET department = 'IT' WHERE department = 'HR';
ROLLBACK TO SAVEPOINT before_update;
COMMIT;
```

In this example, a transaction is started, changes are made and a savepoint is set. If needed, the transaction can be rolled back to the savepoint before being committed.

EXAMPLES

To create a relation:

```
CREATE TABLE employees (  
    id            INTEGER,  
    first_name    CHAR(50),  
    last_name     VARCHAR(100));
```

- To delete a relation:

```
DROP TABLE employees;
```

- To alter a relation (add/remove column): **ALTER TABLE** employees **ADD**

```
age INTEGER; ALTER TABLE employess  
DROP last_name;
```

SQL Values

- Integers and reals are represented as you would expect
 - Strings are too, except they require single quotes.
 - Two single quotes = real quote, e.g., 'Joe''s Bar'
 - Any value can be NULL
 - Unless attribute has NOT NULL constraint
- E.g.: price REAL **NOT NULL**

Dates and Times

- DATE and TIME are types in SQL.
 - The form of a **date value** is: DATE 'yyyy-mm-dd' Example (for Oct. 19, 2011):

DATE '2011-10-19'

- The form of a **time value** is: TIME 'hh:mm:ss' with an optional decimal point and fractions of a second following.
Example (for two and a half seconds after 6:40PM):

TIME '18:40:02.5'

INTEGRITY CONSTRAINTS

Running Example

Beers(name, manf)

Bars(name, addr,
license) Drinkers(name,
addr, phone)

Likes(drinker, beer)

Sells(bar, beer, price)

Frequents(drinker, bar)

Underline = *key* (tuples cannot have the same value in all key attributes)

- Excellent example of a constraint

Kinds of Constraints

- Keys
- Foreign-key or referential-integrity constraints
 - Inter-relation constraints
- Value-based constraints
 - Constrain values of a particular attribute
- Tuple-based constraints
 - Relationship among components
- Assertions

Declaring Keys

- An attribute or list of attributes may be declared **PRIMARY KEY** or **UNIQUE**
 - Either says that no two tuples of the relation may agree in all the attribute(s) on the list
 - There are a few distinctions to be mentioned later
 - Place **PRIMARY KEY** or **UNIQUE** after the type in the declaration of the attribute.

```
CREATE TABLE Beers (  
    name  CHAR(20) PRIMARY KEY,  
    manf  CHAR(20)  
);
```

Foreign Keys

- Values appearing in attributes of one relation must appear together in certain attributes of another relation

Example:

We might expect that a value in **Sells.beer** also appears as value in **Beers.name**

Beers(name, manf)

Sells(bar, beer, price)

Expressing Foreign Keys

- Use keyword **REFERENCES**, either:
 - After an attribute (for one-attribute keys)
REFERENCES <relation> (<attributes>)
 - As an element of the schema:
FOREIGN KEY (<list of attributes>)
REFERENCES <relation> (<attributes>)
- Referenced attributes must be declared **PRIMARY KEY**
or **UNIQUE**

Example: With Attribute

```
CREATE TABLE Beers (  
    name      CHAR(20) PRIMARY  
    KEY, manf  CHAR(20) );
```

```
CREATE TABLE Sells  
    ( bar  
  
    CHAR(20),  
    beer      CHAR(20) REFERENCES  
    Beers(name), price  REAL );
```

Example: As Schema Element

```
CREATE TABLE Beers (  
    name      CHAR(20) PRIMARY  
    KEY, manf  CHAR(20) );
```

```
CREATE TABLE Sells  
    (  
        bar  
        CHAR(20),  
    beer  CHAR(20),  
    price REAL,  
    FOREIGN KEY(beer) REFERENCES Beers(name));
```

Enforcing Foreign-Key Constraints

- If there is a foreign-key constraint from relation **R** to relation **S**, two violations are possible:
 - An insert or update to **R** introduces values not found in **S**
 - A deletion or update to **S** causes some tuples of **R** to “dangle”

Example: suppose **R = Sells**, **S = Beers**

- An insert or update to **Sells** that introduces a non-existent beer
must be rejected
- A deletion or update to **Beers** that removes a beer
value found in some tuples of **Sells** can be handled in

three ways (next slide).

Actions Taken

- **DEFAULT:** Reject the modification
 - Deleted beer in **Beer**: reject modifications in **Sells** tuples
 - Updated beer in **Beer**: reject modifications in **Sells** tuples
- **CASCADE:** Make the same changes in **Sells**
 - Deleted beer in **Beer**: delete **Sells** tuple
 - Updated beer in **Beer**: change value in **Sells**
- **SET NULL:** Change the beer to NULL
 - Deleted beer in **Beer**: set NULL values in **Sells** tuples
 - Updated beer in **Beer**: set NULL values in **Sells** tuples

Example

- Delete the 'Bud' tuple from Beers
 - DEFAULT: do not change any tuple from Sells that have beer = 'Bud'
 - CASCADE: delete all tuples from Sells that have beer = 'Bud'
 - SET NULL: Change all tuples of Sells that have beer = 'Bud' to have beer = NULL
- Update the 'Bud' tuple to 'Budweiser'
 - DEFAULT: do not change any tuple from Sells that have beer = 'Bud'
 - CASCADE: change all Sells tuples with beer = 'Bud' to beer = 'Budweiser'
 - SET NULL: Same change as for deletions

Choosing a Policy

- When we declare a foreign key, we may choose policies **SET NULL** or **CASCADE** independently for deletions and updates
- Follow the foreign-key declaration by:
ON [UPDATE, DELETE][SET NULL, CASCADE]
- Two such clauses may be used, otherwise, the default (reject) is used.

Example: Setting a Policy

```
CREATE TABLE Sells
    ( bar
      beer      CHAR(20),
      price     REAL,
      FOREIGN KEY(beer) REFERENCES Beers(name)
                        ON DELETE SET NULL
                        ON UPDATE CASCADE
    );
```

Attribute-Based Checks

- Constraints on the value of a particular attribute
- Add **CHECK**(<condition>) to the declaration for the attribute
- The condition may use the name of the attribute, but **any other relation or attribute name must be in a subquery**

Example: Attribute-based Check

CREATE TABLE Sells

(bar

CHAR(20),

beer CHAR(20) **CHECK** (beer **IN**(

SELECT name **FROM**

Beers)), price REAL **CHECK** (price <= 5.00)

);

Timing of Checks

- Attribute-based checks are performed **only when** a value for that attribute is inserted or updated
- Example:
 - **CHECK** (price $\leq$ 5.00)
Checks every new price and rejects the modification (for that tuple) if the price is more than \$5

Tuple-Based Checks

- **CHECK** (<condition>) may be added as a relation- schema element
 - The condition may refer to any attribute of the relation, but other attributes or relations require a subquery
 - Checked on insert or update only

Example: Only Joe's Bar can sell beer for more than \$5:

```
CREATE TABLE Sells (  
    bar          CHAR(20),  
    beer         CHAR(20),  
    price        REAL,
```


CHECK (bar = 'Joe''s Bar' OR price <= 5.00));

Assertions

- Permit the definition of constraints over whole tables, rather than individual tuples
 - useful in many situations -- e.g., to express generic inter-relational constraints
 - An assertion associates a name to a check clause. Syntax:
CREATE ASSERTION AssertName **CHECK** (Condition)

Example:

"There must always be at least one tuple in table Employee":

```
CREATE ASSERTION AlwaysOneEmployee  
CHECK (1 <= (SELECT count(*) FROM Employee))
```

Enforcement Policies

- Integrity constraints (checks, assertions) may be checked immediately when a change takes place to a relation, or at the end of a **transaction**
 - The first case may result in a **partial rollback**
 - the latter in a **(full) rollback**.
- This topic is discussed in more detail in **EECS4411**

DATA MANIPULATION LANGUAGE (DML)

Data Manipulation Language (DML)

- Syntax elements used for inserting, deleting and updating data in a database
- Modification statements include:
 - **INSERT** - for inserting data in a database
 - **DELETE** - for deleting data in a database
 - **UPDATE** - for updating data in a database
- All modification statements operate on *a set* of tuples (no duplicates)

Example

Employee(FirstName,Surname,Dept,Office,Salary,City)
Department(DeptName, Address, City)
Product(Code,Name,Description,ProdArea)
LondonProduct(Code,Name,Description)

Insertions

Syntax varies:

- Using only **values**:
INSERT INTO Department
VALUES ('Production', 'Rue du Louvre 23', 'Toulouse')
- Using both **column names** and **values**: **INSERT INTO**
Department(DeptName, City) **VALUES**
('Production', 'Toulouse')
- Using a **subquery**:
INSERT INTO LondonProducts
(**SELECT** Code, Name,

Description **FROM** Product
WHERE ProdArea = 'London')

Notes on Insertions

- The ordering of attributes (if present) and of values is meaningful -- first value for the first attribute, etc.
- If *AttributeList* is omitted, all the relation attributes are considered, in the order they appear in the table definition
- If *AttributeList* does not contain all the relation attributes, left-out attributes are assigned default values (if defined) or the NULL value

Deletions

Syntax:

DELETE FROM *TableName* [**WHERE** Condition]

- "Remove the Production department":
DELETE FROM Department
WHERE DeptName = 'Production'
- "Remove departments with no employees":
DELETE FROM Department
WHERE DeptName **NOT IN** (**SELECT** Dept **FROM** Employee)

Notes on Deletions

- The DELETE statement removes from a table all tuples that satisfy a condition
- If the WHERE clause is omitted, DELETE removes all tuples from the table (keeps the table schema):

DELETE FROM Department

- The removal may produce deletions from other tables – (see referential integrity constraint with **cascade** policy)
- To remove table **Department** completely (content and schema) :

DROP TABLE Department **CASCADE**

Updates

Syntax:

UPDATE *TableName*

SET *Attribute* = < *Expression* | *SelectSQL* | null | default >
{, *Attribute* = < *Expression* | *SelectSQL* | null | default >}
[**WHERE** *Condition*]

- Examples:

UPDATE Employee

SET Salary = Salary + 5

WHERE RegNo = 'M2047'

UPDATE Employee

SET Salary = Salary * 1.1

WHERE Dept = 'Administration'

Notes on Updates

- The order of updates is important:

UPDATE Employee

SET Salary = Salary * 1.15

WHERE Salary <= 30

UPDATE Employee

SET Salary = Salary * 1.1

WHERE Salary > 30

- In this example, some employees may get a double raise
(e.g., employee with salary 29)! How can we fix this?

VIEWS

Views

- A **view** is a relation defined in terms of stored tables (called **base tables**) and other views.

- Two kinds:

- **Virtual** = not stored in the database; just a query for constructing the relation

CREATE VIEW <name> **AS** <query>;

- **Materialized** = actually constructed and stored

CREATE MATERIALIZED VIEW <name> **AS**
<query>;

Running Example

Beers(name, manf)

Bars(name, addr,
license) Drinkers(name,
addr, phone)

Likes(drinker, beer)

Sells(bar, beer, price)

Frequents(drinker, bar)

Example: View Definition

CanDrink(drinker, beer) is a view “containing” the drinker- beer pairs such that the drinker frequents at least one bar that serves the beer:

```
CREATE VIEW CanDrink AS  
  SELECT drinker, beer  
  FROM Frequents, Sells  
  WHERE Frequents.bar = Sells.bar;
```

Example: Accessing a View

Query a view as if it were a base table:

```
SELECT beer  
FROM CanDrink  
WHERE drinker = 'Sally';
```

Notes on Views

- Data independence (hide schema from apps)
 - DB team splits CustomerInfo into Customer and Address
 - View accomodates changes with web apps
- Data hiding (access data on need-to-know basis)
 - Doctor outsources patient billing to third party
 - View restricts access to billing-related patient info
- Code reuse
 - Very similar subquery appears multiple times in a query
 - View shortens code, improves readability, reduces

bugs, ...

- Bonus: query optimizer often does a better job!

Example: Views and Queries

Employee(RegNo,FirstName,Surname,Dept,Office,Salary,City)
Department(DeptName,Address,City)

"Find the department with highest salary expenditures" (**without using a view**):

```
SELECT Dept
FROM Employee
GROUP BY Dept
HAVING sum(Salary) >= ALL (
    SELECT sum(Salary) FROM Employee GROUP BY
    Dept)
```

Example: Views and Queries (cont.)

"Find the department with highest salary expenditures"
(**using a view**):

```
CREATE VIEW SalBudget (Dept, SalTotal) AS  
SELECT Dept, sum(Salary)  
FROM Employee  
GROUP BY Dept
```

```
SELECT Dept  
FROM SalBudget  
WHERE SalTotal = (SELECT max(SalTotal) FROM  
SalBudget)
```

Updates on Views

- Generally, it is impossible to modify a virtual view because it doesn't exist
- Can't we "translate" updates on views into "equivalent" updates on base tables?
 - Not always (in fact, not often)
 - Most systems prohibit most view updates

Example:The View

Pick one copy of
each attribute

CREATE VIEW Synergy

AS SELECT Likes.drinker, Likes.beer,

FROM Likes, Sells,
Frequents

WHERE Likes.drinker =
Frequents.drinker

AND Likes.beer = Sells.beer

Join of Likes, Sells, and Frequents

Interpreting a View Insertion

- We cannot insert into Synergy - it is a virtual view
- **Idea**: Try to translate a (drinker, beer, bar) triple into three insertions of projected pairs, one for each of Likes, Sells, and Frequents.
 - Sells.price will have to be NULL.
 - There isn't always a unique translation

Need for **SQL Triggers** - Not discussed

Materialized Views

- **Problem:** each time a base table changes, the materialized view may change
 - Cannot afford to recompute the view with each change
- **Solution:** Periodic reconstruction of the materialized view, which is otherwise “out of date”

Example: A Data Warehouse

- Wal-Mart stores every sale at every store in a database
- Overnight, the sales for the day are used to update a data warehouse = **materialized views** of the sales
- The warehouse is used by analysts to **predict trends** and move goods to where they are selling best

INDEXES (INDICES)

Index

- **Problem:** needle in haystack
 - Find all phone numbers with first name 'Mary'
 - Find all phone numbers with last name 'Li'
- **Index:** auxiliary database structure which provides random access to data
 - Index a set of attributes. No standard syntax! Typical is:
CREATE INDEX indexName **ON**
TableName(AttributeList);
 - Random access to any indexed attribute
(e.g., retrieve a single tuple out of billions in **<5** disk accesses)
 - Similar to a hash table, but in a DBMS it is a **balanced search tree** with giant nodes (a full disk page) called a *B-tree*

Example: Using Index

```
SELECT fname  
FROM people  
WHERE lname = 'Papagelis'
```

- **Without an index:**
The DBMS must look at the *lname* column on every row in the table (this is known as a **full table scan**)
- **With an index** (defined on attribute *lname*):
The DBMS simply follows the **B-tree** data structure until the 'Papagelis' entry has been found

This is much less computationally expensive than a full table scan

Another Example: Using Index

```
CREATE INDEX BeerInd ON Beers(manf);  
CREATE INDEX SellInd ON Sells(bar, beer);
```

Query: Find the prices of beers manufactured by Pete's and sold by Joe's bar

```
SELECT price FROM Beers, Sells  
WHERE manf = 'Pete''s' AND Beers.name = Sells.beer  
      AND bar = 'Joe''s Bar';
```

DBMS uses:

- **BeerInd** to get all the beers made by Pete's **fast**
- **SellInd** to get prices of those beers, with bar = 'Joe''s Bar' **fast**

What's Next?

- SQL Assignment 2